

Disease Prediction from Symptom Profiles: A Comparative Study of Supervised and Unsupervised Machine Learning

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I. Introduction

Getting a timely and accurate medical diagnosis is still a real problem, especially in places with limited resources. Many healthcare systems rely heavily on doctors being available, and when they're not, people wait or worse, go without answers. Because of that, I see real value in building simple, early-stage tools that can help flag possible conditions based on symptoms.

In this project, I aim to build and compare machine learning models that can predict likely diseases from patient symptoms. The idea is not to replace doctors, but to create something that can assist in early screening and decision-making. I'll be using a publicly available dataset from Kaggle that includes symptom and patient profile information.

The core problem is a multi-class classification task: given a set of observed symptoms and demographic features for a patient, identify which of several diseases is most likely present. Beyond classification, the project extends into unsupervised learning by applying clustering algorithms to the same dataset, examining whether natural groupings emerge from symptom patterns without the use of disease labels. Comparing these two paradigms offers insight into both the structure of the data and the relative utility of label-driven versus discovery-driven approaches in a healthcare context.

This project is personal to me. From 2009 to 2016, I helped community medical clinics in Indonesia, where limited physician access meant that many patients went without timely diagnoses. Building something that can even partially help identify potential illnesses from symptoms is not just a technical exercise for me, it's something I've wanted to do for a long time.

II. Related Work

A lot of past work has already explored using machine learning to predict diseases from patient data, and I'm building on those ideas rather than starting from scratch.

One approach that stands out is K-Nearest Neighbors (KNN). Research like Guo et al. (2003) shows that KNN works well for medical datasets because it focuses on similarity, patients with similar symptoms are likely to have similar conditions. This makes sense, especially when the relationship between symptoms and diseases isn't perfectly linear.

Another strong approach is ensemble methods, especially Random Forest, introduced by Breiman (2001). Instead of relying on a single model, Random Forest combines multiple

decision trees, which helps handle complex relationships in the data and reduces overfitting. It's also known to perform well when features are correlated, which is common in medical data where symptoms often appear together.

On the unsupervised side, clustering methods like K-means and agglomerative clustering have been used to group patients without using disease labels. Work by Xu & Wunsch (2005) shows that these methods can reveal hidden structures in the data, like patterns of symptom co-occurrence or natural groupings of patients.

In this project, I'm bringing these ideas together. I'll use Logistic Regression as a simple and interpretable baseline, Random Forest as a stronger supervised model, and KNN as a comparison point. Then, I'll apply K-means and agglomerative clustering to see what patterns show up when the model isn't given any labels. The goal is not just performance, but understanding how these different approaches behave on the same problem.

III. Proposed Work

I'm structuring this project in four main stages: preprocessing the data, building supervised models, applying unsupervised methods, and then comparing the results.

Dataset.

I'll be using the Kaggle Disease Symptoms and Patient Profile dataset. It includes patient records with binary symptom indicators like fever, cough, and fatigue, along with demographic features such as age, gender, blood pressure, and cholesterol. Each record is labeled with a disease, which makes it suitable for supervised learning, but I can also remove the labels for clustering.

Preprocessing.

Before building any models, I'll clean and prepare the data. This includes checking for missing values and either imputing or removing them depending on the situation. I'll encode categorical variables using label encoding and normalize the data using Min-Max scaling, especially for models that depend on distance like KNN and clustering. I'll also look at the class distribution to understand if there's any imbalance.

Supervised Learning.

I'll train three models: Logistic Regression, KNN, and Random Forest. Logistic Regression gives me a baseline that's easy to interpret. KNN helps capture similarity-based patterns, and Random Forest will likely give the strongest performance. I'll tune hyperparameters where needed, especially for KNN. Evaluation will be done using accuracy, precision, recall, F1 score, and confusion matrices, using an 80/20 train-test split.

Unsupervised Learning.

For the unsupervised part, I'll apply K-means and agglomerative clustering to the dataset without the disease labels. I'll set the number of clusters equal to the number of disease classes so I can compare them later. To evaluate how well the clusters match the actual diseases, I'll use Normalized Mutual Information (NMI). I'll also use PCA to reduce the data to two dimensions for visualization, so I can actually see how the clusters are formed.

Comparative Analysis.

In the final step, I'll compare everything. I want to see whether clustering methods can naturally recover disease groupings, and how close they get compared to supervised models. I'll also compare the supervised models themselves, looking at which one performs best, and which one is easier to interpret. The results will be summarized using tables and visualizations.

IV. Timeline

Milestone	Description	Due Date
Proposal	Topic, motivation, approach	March 27, 2026
Progress Report I	Dataset loaded, EDA complete	April 6, 2026
Progress Report II	Models trained, initial results	April 13, 2026
Lightning Talk	Oral presentation of findings	April 21–23, 2026
Final Report	Full written report submitted	April 27, 2026

V. References

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.

Guo, G., Wang, H., Bell, D., Bi, Y., & Greer, K. (2003). KNN model-based approach in classification. In *Proceedings of OTM Confederated International Conferences*.

Xu, R., & Wunsch, D. (2005). Survey of clustering algorithms. *IEEE Transactions on Neural Networks*, 16(3), 645–678.

Kaggle. Disease Symptoms and Patient Profile Dataset.

<https://www.kaggle.com/datasets/uom190346a/disease-symptoms-and-patient-profile-dataset>